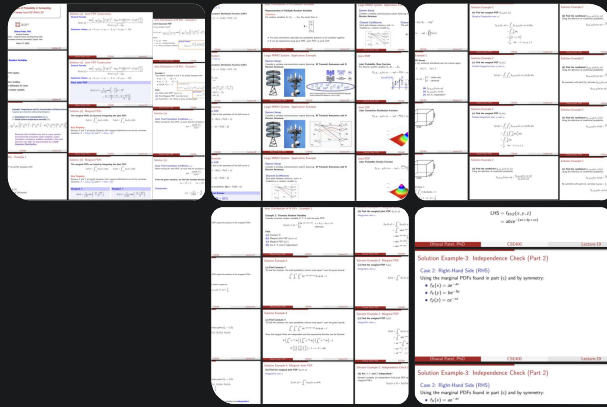


Lecture Scribe Joint Distribution



Create a lecture scribe that will be used as a reference to a good preparation of the exams.

Take the given context (lecture material and other material in the textbook). The content of the output should be very strict to the material presented. Introduce no new explanations, examples, results or interpretations not explicitly set forth in context.

Content Requirements

Included should be that material that is present in the given context like:

1. Definitions and notation
2. Assumptions and declared conditions.
3. Formulas, outcomes, and theories.
4. Proofs or derivations shown
5. The worked examples and solved problems.
6. In between steps depicted in solutions.
7. Rational progression among ideas as they exist.

Scope Constraints

1. Only given lecture and textbook materials.
2. No additional issues, alternative solutions, intuition or commentary are to be added unless indicated.
3. Not to follow the steps that are in the source material.
4. Be consistent in the notation and terminology.

Structural Requirements

1. Ascertain the scribe in sequence of presentation in order of appearance using systematic parts like:
2. Title
3. Definitions and Notation
4. Theorems or Results
5. Proofs or Derivations

6. Solved Problems or Worked Examples.

7. Maintain the initial order of subjects.

Objective

Create a faithful, structured lecture scribe that would allow the correct revision of definitions and derivations and solved problems by use of the given document alone.

Lecture Scribe: Joint Distribution of N Random Variables (CSE400 – Lecture 19)

1. Definitions and Notation

Joint Distribution of N Random Variables

- For random variables $X_1, X_2, \dots, X_N$, the vector form is:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- The joint distribution describes the probability behavior of all variables together.
 - It can be represented using:
 - Joint PMF
 - Joint PDF
 - Joint CDF
-

Joint PMF (Discrete Case)

For discrete random variables:

$$P_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = P(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N)$$

Joint CDF

$$F_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_N \leq x_N)$$

Joint PDF

$$f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = \frac{\partial^N F_{X_1, X_2, \dots, X_N}}{\partial x_1 \partial x_2 \cdots \partial x_N}$$

2. Jointly Gaussian Random Variables

Joint Gaussian PDF (Two Variables)

For X and Y :

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

Where:

- μ_X, μ_Y : means
 - σ_X, σ_Y : standard deviations
 - ρ : correlation coefficient
-

Correlation Coefficient

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X\sigma_Y}$$

3. Marginal Distributions

- Marginals are obtained by integrating the joint PDF:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

Key Property (Gaussian Case)

If X and Y are jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

4. Independence of Random Variables

Condition for Independence

Random variables are independent if:

$$f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

5. Worked Examples

Example 1: Joint Gaussian Distribution**Given**

- $\mu_X = 2, \sigma_X^2 = 1 \Rightarrow \sigma_X = 1$
 - $\mu_Y = -3, \sigma_Y^2 = 4 \Rightarrow \sigma_Y = 2$
 - $\text{cov}(X, Y) = -1$
-

(a) Find Correlation Coefficient

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

(b) Joint PDF

Substitute values into Gaussian formula:

$$f_{X,Y}(x, y) = \frac{1}{2\pi(1)(2)\sqrt{1 - (\frac{1}{2})^2}} \exp \left(-\frac{1}{2(1 - \frac{1}{4})} \left[(x - 2)^2 + \left(\frac{y + 3}{2} \right)^2 - 2 \left(-\frac{1}{2} \right) (x - 2) \left(\frac{y + 3}{2} \right) \right] \right)$$

Final simplified form:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp \left(-\frac{2}{3} \left[(x - 2)^2 + \frac{(x - 2)(y + 3)}{2} + \frac{(y + 3)^2}{4} \right] \right)$$

(c) Marginal PDFs

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{(x - 2)^2}{2} \right)$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp \left(-\frac{(y + 3)^2}{8} \right)$$

(d) Find $P(X < 0)$ **Step 1: Standardization**

$$Z = \frac{X - \mu_X}{\sigma_X} = \frac{X - 2}{1}$$

$$P(X < 0) = P(Z < -2)$$

Step 2: Symmetry Property

$$P(Z < -2) = P(Z > 2)$$

Step 3: Q-function

$$P(X < 0) = Q(2)$$

6. Example 2: Uniformly Distributed 3D Vector

Given

$$|x_1| \leq 1, \quad |x_2| \leq 1, \quad |x_3| \leq 1$$

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} C, & \text{inside cube} \\ 0, & \text{otherwise} \end{cases}$$

(a) Find Constant C

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C \, dx_1 dx_2 dx_3 = 1$$

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

(b) Marginal $f_{X_1, X_2}(x_1, x_2)$

$$f_{X_1, X_2}(x_1, x_2) = \int_{-1}^1 \frac{1}{8} dx_3 = \frac{1}{8}(2) = \frac{1}{4}$$

(c) Marginal $f_{X_1}(x_1)$

$$f_{X_1}(x_1) = \int_{-1}^1 \int_{-1}^1 \frac{1}{8} dx_2 dx_3 = \frac{1}{8}(2)(2) = \frac{1}{2}$$

(d) Conditional PDF

$$f_{X_1|X_2, X_3}(x_1|x_2, x_3) = \frac{f_{X_1, X_2, X_3}(x_1, x_2, x_3)}{f_{X_2, X_3}(x_2, x_3)} = \frac{1/8}{1/4} = \frac{1}{2}$$

(e) Conditional PDF

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_2}(x_2)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(f) Independence Check

LHS:

$$f_{X_1, X_2, X_3} = \frac{1}{8}$$

RHS:

$$f_{X_1}(x_1)f_{X_2}(x_2)f_{X_3}(x_3) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion:

$$\text{LHS} = \text{RHS} \Rightarrow \text{independent}$$

7. Example 3: Trivariate Distribution

Given

$$f_{X,Y,Z}(x,y,z) = \begin{cases} Ke^{-(ax+by+cz)}, & x, y, z > 0 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find Constant K

$$\int_0^\infty \int_0^\infty \int_0^\infty Ke^{-(ax+by+cz)} dx dy dz = 1$$

$$K \left(\frac{1}{a}\right) \left(\frac{1}{b}\right) \left(\frac{1}{c}\right) = 1 \Rightarrow K = abc$$

(b) Marginal Joint PDF $f_{X,Y}(x,y)$

$$f_{X,Y}(x,y) = \int_0^\infty abce^{-(ax+by+cz)} dz = abe^{-(ax+by)}$$

(c) Marginal PDF $f_X(x)$

$$f_X(x) = \int_0^\infty ae^{-ax}be^{-by}dy = ae^{-ax}, \quad x > 0$$

(d) Independence Check

LHS:

$$f_{X,Y,Z}(x,y,z) = abce^{-(ax+by+cz)}$$

RHS:

$$f_X(x)f_Y(y)f_Z(z) = (ae^{-ax})(be^{-by})(ce^{-cz}) = abce^{-(ax+by+cz)}$$

Conclusion:

$$\text{LHS} = \text{RHS} \Rightarrow X, Y, Z \text{ are independent}$$

8. Application: Large MIMO System

System Setup

- M : Transmit antennas
 - N : Receive antennas
-

Channel Coefficients

- Each path between transmit antenna j and receive antenna i :

$$h_{ij}$$

Channel Matrix

$$H = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1N} \\ h_{21} & h_{22} & \cdots & h_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ h_{M1} & h_{M2} & \cdots & h_{MN} \end{bmatrix}$$

End of Lecture Scribe